

CUNY SPS
DATA 620 Web Analytics
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Week Four Assignment: **Centrality Measures**

Analysis of Centrality Across Twitch Partner Status

Data Source and Network Selection

For this week's assignment, I opted for the English-language Twitch Social Network dataset made available through the Stanford Network Analysis Project (SNAP, n.d.). The dataset forms part of a larger collection of Twitch social networks encompassing users across six language-specific communities: English, German, French, Spanish, Portuguese, and Russian. For the purposes of the proposed analysis, I elected to focus specifically upon the English-language network, which consists of 7,126 nodes and 35,324 edges (SNAP, n.d.). Within the network, each node represents a Twitch user, whilst the edges represent mutual friendship relationships between users. The dataset was originally compiled as part of the work conducted by Rozemberczki et al. (2021) on attributed social network data.

My selection of Twitch was driven, in part, by a relatively recent personal interest in the streaming medium. In my free time, I happened upon a streamer broadcasting whilst travelling throughout South Korea and found the experience surprisingly immersive. Rather than simply watching someone else's travels, the stream provided what I would characterize as a form of augmented exploration, wherein I could experience the streets, neighborhoods, and everyday environment of Seoul from the perspective of the individual physically navigating them. This consequently prompted a broader interest in streaming as a medium and, particularly for the purposes of this assignment, the social networks that exist among its creators.

The downloaded English-language dataset contains an edge list alongside information pertaining to the individual nodes. Of particular interest is the partner variable, which categorizes each streamer according to whether or not they possessed Twitch Partner status within the dataset. Twitch Partner status is associated with Twitch's Partner Program, a designation through which qualifying creators gain access to additional opportunities and benefits on the platform (Twitch, 2026). This provides a categorical distinction between Partner (True) and non-Partner (False) streamers that can subsequently be compared with measures of network centrality. Moreover, the node-level data contain additional information such as the number of views associated with each streamer and the age of each account in days at the time of data collection, thereby providing opportunities for supplementary analysis of how network positioning may relate to other streamer characteristics (Rozemberczki et al., 2021).

Proposed Analysis

The proposed analysis would begin by downloading the English-language Twitch network data from the Stanford Network Analysis Project (SNAP) and subsequently uploading the required CSV files to the project's GitHub repository. From there, the raw GitHub URLs could be utilized to load the data directly into a Google Colab notebook using pandas. The `musae_ENGB_edges.csv` file, which contains the from and to node identifiers, would then be utilized with NetworkX to construct the Twitch social network. The accompanying `musae_ENGB_target.csv` file contains the attributes associated with each streamer, including views, account age in days, and categorical variables such as partner. Importantly, the `new_id` variable contained within this file corresponds to the numerical node identifiers utilized within the edge list. Consequently, the two datasets could be associated through these identifiers, allowing each node within the constructed network to be linked with its corresponding Partner status and other available attributes.

Degree centrality would then be calculated for each streamer in order to assess the extent of their direct connectivity within the network. The resulting centrality measures could subsequently be compared across the Partner and non-Partner categorical groups. Descriptive statistics and visualizations could be utilized to examine whether one group tends to possess greater degree centrality, whilst an appropriate statistical test could later be employed to assess whether any observed difference between the groups is statistically meaningful.

Beyond this principal comparison, the structure of the network presents several additional avenues of interest. For instance, the analysis could examine whether partnered streamers demonstrate a tendency to connect with other partnered streamers or whether their connections are distributed more broadly across Partner and non-Partner nodes. Conversely, it would be interesting to examine whether non-partnered streamers tend to be concentrated within their own portions of the network, potentially indicating some degree of separation between the two categorical groups. Similarly, the available views and days variables could provide additional context regarding the characteristics of the streamers occupying these network positions. One particular avenue of interest may concern partnered streamers possessing newer accounts. The analysis could examine whether these newer Partners are positioned near clusters of more established partnered streamers possessing higher recorded view counts and/or older accounts on the platform. Such a pattern could raise thought-provoking questions regarding whether proximity to established Partners, and the potential network influence associated with such connections, may be related to certain streamers possessing Partner status fairly early in the lifespan of their accounts.

Hypothetical Outcome

A hypothetical outcome of the proposed analysis is that partnered Twitch streamers may exhibit higher average degree centrality than non-partnered streamers. Such a result would indicate that, within the observed English-language Twitch network, partnered streamers tend to maintain a greater number of direct connections with other users. In this regard, degree

centrality could potentially provide some predictive information regarding whether a particular node belongs to the Partner or non-Partner categorical group.

Furthermore, should partnered streamers exhibit greater degree centrality, an examination of the additional node attributes could provide greater context for this distinction. For example, Partners may also tend to possess higher recorded view counts or older accounts within the dataset. Conversely, identifying partnered streamers with relatively fewer days but close network proximity to other Partners could present an interesting avenue for examining whether network positioning differs between newer Partners and their more established counterparts.

However, any such association would not, in itself, establish that greater network centrality or proximity to established Partners causes a streamer to attain Partner status. Other characteristics, including account age, audience size, streaming activity, and the broader position of an individual within the network, may also relate to both connectivity and Partner status. Consequently, the proposed comparison could serve as an initial means of assessing whether a discernible relationship exists between a streamer's direct network connectivity and their Partner classification, whilst the additional node attributes provide opportunities for further examination of the characteristics that may accompany such a relationship.

References

- Rozemberczki, B., Allen, C., & Sarkar, R. (2021). Multi-scale attributed node embedding. *Journal of Complex Networks*, 9(2), cnab014. <https://doi.org/10.1093/comnet/cnab014>
- Stanford Network Analysis Project. (n.d.). *Twitch social networks*. Stanford University. <https://snap.stanford.edu/data/twitch-social-networks.html>
- Twitch. (2026, May 13). *Monetization for all*. Twitch Blog. <https://blog.twitch.tv/en/2026/05/13/monetization-for-all/>